

MATH3900: Optimization for AI and DA

Unconstrained Optimization

Prof. Lexiao Lai
Department of Mathematics
The University of Hong Kong

Semester 1, 2026–2027



Unconstrained minimization problems

- ▶ Consider the problem:

$$\min_{x \in \mathbb{R}^d} f(x),$$

- ▶ where $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable.
- ▶ For logistic regression,

$$f(x) := \frac{1}{n} \sum_{i=1}^n \log \left(1 + e^{-y_i a_i^T x} \right).$$

- ▶ In general, there is no closed-form formula for a minimizer.
- ▶ We need an iterative algorithm to *approximate* it:
 1. In which direction should we move?
 2. How far should we move?
 3. When can we stop, and how can we tell whether the algorithm returns a good enough solution?

- ▶ How can we tell whether a candidate point is a minimizer?
- ▶ The local behavior of a smooth function is given by its derivatives.
- ▶ For a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, a point $x^* \in \mathbb{R}^d$ is **stationary** if $\nabla f(x^*) = 0$.

Theorem

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable at x^* . If x^* is a local minimizer, then

$$\nabla f(x^*) = 0.$$

- ▶ For an unconstrained problem, this is a **first-order necessary optimality condition**: every differentiable local minimizer must satisfy it.

Does stationarity imply optimality?

- ▶ A necessary condition filters candidates, but it need not identify a minimizer. At $x = 0$:

$f(x) = x^2$	$f(x) = -x^2$	$f(x) = x^3$
$f'(0) = 0$	$f'(0) = 0$	$f'(0) = 0$
local minimizer	local maximizer	neither

- ▶ All three points are stationary, but only the first is a local minimizer.
- ▶ How can we tell whether a stationary point is a minimizer?
- ▶ Answer: Examine higher-order derivatives, which measure *curvature*.

Positive semidefinite matrices

- ▶ Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a twice continuously differentiable function; its Hessian $\nabla^2 f(x) \in \mathbb{R}^{d \times d}$ is symmetric.
- ▶ How do we denote positivity for matrices?

Definition

Let $Q \in \mathbb{R}^{d \times d}$ be a symmetric matrix, i.e., $Q = Q^T$.

- ▶ Q is **positive semidefinite** if $z^T Q z \geq 0$ for every $z \in \mathbb{R}^d$.
- ▶ Q is **positive definite** if $z^T Q z > 0$ for every nonzero $z \in \mathbb{R}^d$.
- ▶ Shorthands: $Q \succeq 0$ for positive semidefinite and $Q \succ 0$ for positive definite.
- ▶ For symmetric P, Q , write $P \preceq Q$ when $Q - P \succeq 0$.

Theorem

Every symmetric matrix $Q \in \mathbb{R}^{d \times d}$ admits a decomposition

$$Q = U\Lambda U^T, \quad U^T U = U U^T = I, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d),$$

where the eigenvalues $\lambda_1, \dots, \lambda_d$ are real.

- Order the eigenvalues so that $\lambda_{\min}(Q) = \lambda_1 \leq \dots \leq \lambda_d = \lambda_{\max}(Q)$.

Theorem

For every symmetric matrix $Q \in \mathbb{R}^{d \times d}$,

$$\lambda_{\min}(Q)I \preceq Q \preceq \lambda_{\max}(Q)I.$$

It follows that

- $Q \succeq 0$ iff all eigenvalues are nonnegative, and
- $Q \succ 0$ iff all eigenvalues are positive.

Theorem

Suppose f is twice continuously differentiable near x^* .

- ▶ **Necessary:** if x^* is a local minimizer, then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$.
- ▶ **Sufficient:** if $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$, then x^* is a local minimizer.

Which direction should we move?

- ▶ Assume that we are at a nonoptimal point x_k . Which direction should we move?
- ▶ Idea: move in a direction that decreases the objective most rapidly.
- ▶ By the first-order Taylor expansion, as $t \downarrow 0$,

$$f(x_k + tp) = f(x_k) + t\nabla f(x_k)^T p + o(t).$$

- ▶ Thus p is a **descent direction** when $\nabla f(x_k)^T p < 0$.
- ▶ What is the steepest descent direction?

Theorem: Cauchy–Schwarz inequality

For any $u, v \in \mathbb{R}^d$, $|u^T v| \leq \|u\| \|v\|$.

For every p with $\|p\| = 1$,

$$\nabla f(x_k)^T p \geq -\|\nabla f(x_k)\|.$$

If $\nabla f(x_k) \neq 0$, equality is attained by

$$p_k = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}.$$

Thus $-\nabla f(x_k)$ is the steepest descent direction.

Algorithm (Gradient descent)

Require: $x^0 \in \mathbb{R}^d$, $\varepsilon \geq 0$, and a step-size rule to determine (α_k)

```
1: for  $k = 0, 1, 2, \dots$  do  
2:   if  $\|\nabla f(x^k)\| \leq \varepsilon$  then return  $x^k$   
3:   Choose  $\alpha_k > 0$  using the step-size rule  
4:    $x^{k+1} \leftarrow x^k - \alpha_k \nabla f(x^k)$   
5: end for
```

- ▶ The gradient selects the direction; the step size controls the distance traveled.
- ▶ Example: For $f(x) = \frac{1}{2}(9x_1^2 + x_2^2)$, start at $x^0 = (1, 1)^T$ and take $\alpha_0 = 0.1$. What is the next iterate x^1 ?

How should we choose step sizes?

- ▶ For the simplest quadratic $f(x) = \frac{1}{2}x^2$, with $x_0 \neq 0$, gradient descent is $x_{k+1} = (1 - \alpha)x_k$.

Step size	Multiplier	Behavior
$0 < \alpha < 2$	$ 1 - \alpha < 1$	converges to 0
$\alpha = 2$	$1 - \alpha = -1$	oscillates forever
$\alpha > 2$	$ 1 - \alpha > 1$	diverges in magnitude

- ▶ A poor step size can prevent convergence.
- ▶ A safe step must be small relative to how quickly the gradient changes.

Definition

Let $D \subseteq \mathbb{R}^d$. A map $G : D \rightarrow \mathbb{R}^m$ is **L -Lipschitz** on D , with $L \geq 0$, if

$$\|G(u) - G(v)\| \leq L \|u - v\| \quad \text{for all } u, v \in D.$$

- ▶ The output changes by at most L times the change in the input.
- ▶ A differentiable $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **L -smooth** when $G = \nabla f$ is L -Lipschitz:

$$\|\nabla f(u) - \nabla f(v)\| \leq L \|u - v\| \quad \text{for all } u, v \in \mathbb{R}^d.$$

- ▶ Thus L limits how quickly the gradient can change.

Theorem

If f is L -smooth, then

$$f(v) \leq f(u) + \nabla f(u)^T (v - u) + \frac{L}{2} \|v - u\|^2.$$

- ▶ Gradient descent decreases f when $0 < \alpha < 2/L$.
- ▶ The convenient choice $\alpha = 1/L$ gives $f(x_{k+1}) \leq f(x_k) - (2L)^{-1} \|\nabla f(x_k)\|^2$.

Theorem

Assume f is bounded below and L -smooth for some $L > 0$. Run gradient descent with $\varepsilon = 0$ and $\alpha_k = 1/L$. For every $k \geq 1$,

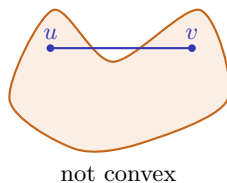
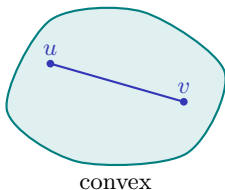
$$\min_{0 \leq j < k} \|\nabla f(x_j)\|^2 \leq \frac{2L(f(x_0) - f^*)}{k}, \quad f^* := \inf_x f(x).$$

- ▶ At least one of the first k iterates has a small gradient.
- ▶ A small gradient does not, by itself, certify a local or global minimizer.
- ▶ Under what condition is a stronger conclusion attainable?

Definition

A set $C \subseteq \mathbb{R}^d$ is **convex** if

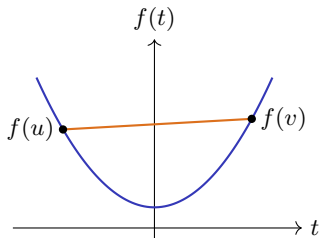
$$u, v \in C, \quad \theta \in [0, 1] \quad \implies \quad \theta u + (1 - \theta)v \in C.$$



Definition

Let C be convex. A function $f : C \rightarrow \mathbb{R}$ is **convex** if, for every $u, v \in C$ and $\theta \in [0, 1]$,

$$f(\theta u + (1 - \theta)v) \leq \theta f(u) + (1 - \theta)f(v).$$



- ▶ The graph lies below every line segment joining two graph points.
- ▶ Equivalently, the epigraph (region above the graph) is convex.

Theorem

The following operations preserve convexity.

- ▶ **Nonnegative sums.** If $f_1, \dots, f_r : \mathbb{R}^d \rightarrow \mathbb{R}$ are convex and $\gamma_1, \dots, \gamma_r \geq 0$, then $\sum_{i=1}^r \gamma_i f_i$ is convex.
 - ▶ **Affine composition.** If $g : \mathbb{R}^m \rightarrow \mathbb{R}$ is convex, $M \in \mathbb{R}^{m \times d}$, and $q \in \mathbb{R}^m$, then $x \mapsto g(Mx + q)$ is convex.
 - ▶ **Pointwise maximum.** If $f_1, \dots, f_r : \mathbb{R}^d \rightarrow \mathbb{R}$ are convex, then $x \mapsto \max_{1 \leq i \leq r} f_i(x)$ is convex.
- ▶ These rules let us build convex objectives from simpler convex pieces.

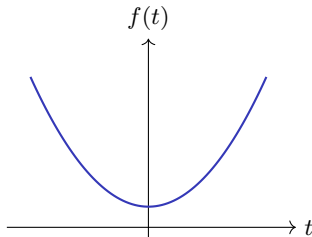
First-order test for convexity

Theorem

Let $C \subseteq \mathbb{R}^d$ be open and convex, and let $f : C \rightarrow \mathbb{R}$ be differentiable. Then f is convex if and only if

$$f(v) \geq f(u) + \nabla f(u)^T (v - u) \quad \text{for all } u, v \in C.$$

- For convex f , every affine tangent model is a **global lower bound**.



Corollary

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable and convex. Then

$$x^* \text{ is a global minimizer} \iff \nabla f(x^*) = 0.$$

Theorem

Let $C \subseteq \mathbb{R}^d$ be open and convex, and let $f : C \rightarrow \mathbb{R}$ be twice continuously differentiable. Then

$$f \text{ is convex on } C \iff \nabla^2 f(x) \succeq 0 \text{ for every } x \in C.$$

Examples:

- ▶ Under what condition is the quadratic function $f(x) := \frac{1}{2}x^T Qx - b^T x$ convex?
- ▶ For logistic regression, recall

$$F(x) = \frac{1}{n} \sum_{i=1}^n \log\left(1 + e^{-y_i a_i^T x}\right).$$

Is this objective convex?

Convexity of logistic-regression objective

- ▶ Consider the objective

$$F(x) = \frac{1}{n} \sum_{i=1}^n \log\left(1 + e^{-y_i a_i^T x}\right).$$

- ▶ For sample i , let $m_i(x) = y_i a_i^T x$ and $\ell_i(x) = \log(1 + e^{-m_i(x)})$. Define $w_i(x) = 1/(1 + e^{m_i(x)})$.
- ▶ Its gradient and Hessian are

$$\nabla \ell_i(x) = -y_i w_i(x) a_i, \quad \nabla^2 \ell_i(x) = w_i(x)(1 - w_i(x)) a_i a_i^T.$$

- ▶ Averaging over the samples gives

$$\nabla^2 F(x) = \frac{1}{n} \sum_{i=1}^n w_i(x)(1 - w_i(x)) a_i a_i^T \succeq 0.$$

- ▶ Since $w_i(x)(1 - w_i(x)) \geq 0$ and $a_i a_i^T \succeq 0$, the Hessian test shows that F is convex.

Convergence of gradient descent on convex functions

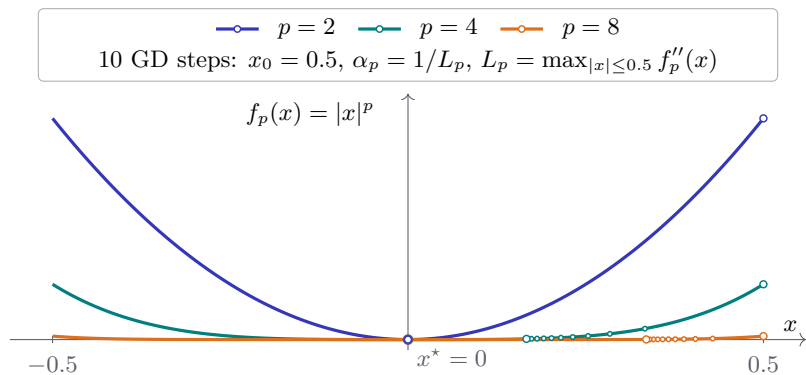
Theorem

Let f be convex and L -smooth for some $L > 0$, and let x^* be a minimizer. Run gradient descent with $\varepsilon = 0$ and $\alpha_k = 1/L$. For every $k \geq 1$,

$$f(x_k) - f(x^*) \leq \frac{L \|x_0 - x^*\|^2}{2k} = O(k^{-1}).$$

- For $\delta > 0$, this bound guarantees an objective gap at most δ after $O(1/\delta)$ iterations.

Why can gradient descent slow down near a minimizer?



- ▶ Gradient descent slows near a flat minimizer.
- ▶ How can we characterize this flatness?
- ▶ At a flat minimizer, the Hessian is degenerate ($\lambda_{\min}(\nabla^2 f(x^*)) = 0$), so the gradient varies slowly nearby.
- ▶ Does gradient descent converge faster near a non-flat minimizer?

Definition

Let $\mu > 0$. A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is **μ -strongly convex** if

$$f(v) \geq f(u) + \nabla f(u)^T(v - u) + \frac{\mu}{2} \|v - u\|^2 \quad \text{for all } u, v \in \mathbb{R}^d.$$

- ▶ A strongly convex function is lower bounded globally by a quadratic function.

Theorem

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable, and let $0 < \mu \leq L$. Then f is μ -strongly convex and L -smooth if and only if

$$\mu I \preceq \nabla^2 f(x) \preceq LI \quad \text{for every } x \in \mathbb{R}^d.$$

- ▶ $\kappa := L/\mu \geq 1$ is called the **condition number**.
- ▶ A problem is ill-conditioned if κ is large.

Theorem

Let f be μ -strongly convex and L -smooth, and let x^* be its unique minimizer. Run gradient descent with $\varepsilon = 0$ and $\alpha_k = 1/L$. For every $k \geq 0$,

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f(x^*)).$$

- ▶ The objective gap decays at a geometric (linear) rate.
- ▶ This rate is much faster than $O(1/k)$ for the convex case.
- ▶ Since $\mu/L = 1/\kappa$, reducing the objective gap by a factor $\delta \in (0, 1)$ takes $O(\kappa \log(1/\delta))$ iterations.
- ▶ What does a problem with a large condition number look like, and why does it slow gradient descent?

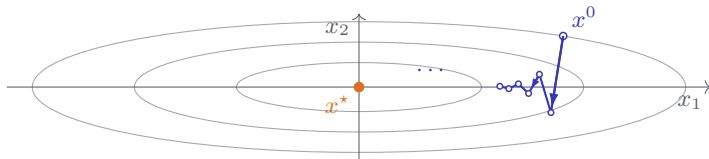
What does a large condition number look like?

Curvature of convex quadratics

For $f(x) = \frac{1}{2}x^T Qx - b^T x$, where $Q = Q^T \succ 0$, we have $\nabla f(x) = Qx - b$ and $\nabla^2 f(x) = Q$. The sharp curvature constants are

$$\mu = \lambda_{\min}(Q), \quad L = \lambda_{\max}(Q), \quad \kappa(f) = \frac{L}{\mu} = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}.$$

$$f(x) = \frac{1}{2}(x_1^2 + 25x_2^2), \quad \kappa = 25, \quad x^0 = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}, \quad \alpha = \frac{3}{50}.$$



- The magnitude of x_2 shrinks quickly, while x_1 moves slowly along the flat direction.

Why do the coordinates behave differently?

- ▶ Consider the d -dimensional diagonal quadratic

$$f(x) = \frac{1}{2}x^T Qx, \quad Q = \text{diag}(\lambda_1, \dots, \lambda_d),$$
$$0 < \mu = \lambda_1 \leq \dots \leq \lambda_d = L.$$

- ▶ For each $i = 1, \dots, d$,

$$x_i^{k+1} = x_i^k - \alpha \lambda_i x_i^k.$$

- ▶ Each coordinate evolves independently, with the common step size α and a multiplier determined by its curvature λ_i .
- ▶ With $\alpha = 1/L$, the slowest coordinate x_1 contracts by the slow factor $1 - \mu/L$.
- ▶ Can we scale the update of each coordinate using its own curvature?

Recall: Taylor models and gradient descent

- ▶ Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable. The gradient descent direction is obtained by examining the first-order Taylor approximation:

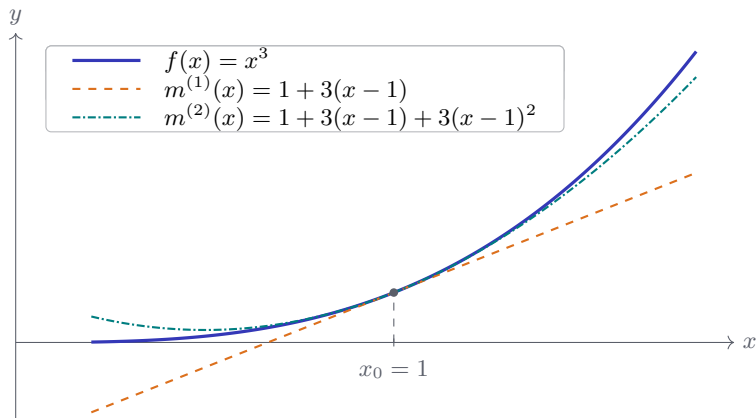
$$f(x_k + p) = \underbrace{f(x_k) + \nabla f(x_k)^T p}_{m_k^{(1)}(p)} + o(\|p\|).$$

- ▶ Recall that we also have

$$f(x_k + p) = \underbrace{f(x_k) + \nabla f(x_k)^T p + \frac{1}{2} p^T \nabla^2 f(x_k) p}_{m_k^{(2)}(p)} + o(\|p\|^2).$$

- ▶ When the Hessian is available, $m_k^{(2)}$ is a more accurate local model because it also matches curvature.

How do first- and second-order approximations compare?



- ▶ $m^{(1)}$ matches the value and slope at x_0 ;
- ▶ $m^{(2)}$ also matches the curvature and follows f more closely near x_0 .
- ▶ Can we obtain the next iterate by minimizing the second-order model?

Which direction does the second-order model suggest?

- ▶ The closer local fit motivates choosing the step from the second-order Taylor model:

$$m_k^{(2)}(p) = f(x_k) + \nabla f(x_k)^T p + \frac{1}{2} p^T \nabla^2 f(x_k) p,$$

where $H_k = \nabla^2 f(x_k)$.

Algorithm (Newton's method)

At each iteration, solve the stationarity equation for the model:

$$\nabla m_k^{(2)}(p_k) = \nabla f(x_k) + H_k p_k = 0, \quad x_{k+1} = x_k + p_k.$$

- ▶ If $H_k \succ 0$, then $p_k = -H_k^{-1} \nabla f(x_k)$ minimizes the quadratic model.
- ▶ Solve $H_k p_k = -\nabla f(x_k)$ directly; do not compute the inverse H_k^{-1} .

Newton's method on a quadratic

- ▶ Let $f(x) = \frac{1}{2}x^T Qx - b^T x$ with $Q \succ 0$. What is the update of Newton's method?
- ▶ $\nabla f(x_k) = Qx_k - b$ and $H_k = Q$.
- ▶ The Newton equation is $Qp_k = -(Qx_k - b)$, so

$$\begin{aligned}x_{k+1} &= x_k - Q^{-1}(Qx_k - b) \\ &= Q^{-1}b = x^*.\end{aligned}$$

- ▶ Because a quadratic equals its second-order Taylor model, exact Newton reaches the minimizer in one iteration.

What controls the change in curvature?

Definition

For $A \in \mathbb{R}^{m \times d}$, its (induced) **matrix 2-norm** is

$$\|A\| := \max_{\|z\|=1} \|Az\|.$$

Consequently, $\|Az\| \leq \|A\| \|z\|$ for every $z \in \mathbb{R}^d$.

- ▶ If $A \in \mathbb{R}^{d \times d}$ is symmetric, then $\|A\| = \max_i |\lambda_i(A)|$. In particular, if $A \succeq 0$, then $\|A\| = \lambda_{\max}(A)$.

Definition

The Hessian is **ρ -Lipschitz** on $D \subseteq \mathbb{R}^d$, with $\rho \geq 0$, if

$$\|\nabla^2 f(u) - \nabla^2 f(v)\| \leq \rho \|u - v\| \quad \text{for all } u, v \in D.$$

- ▶ A quadratic has a constant Hessian, so it satisfies this with $\rho = 0$.
- ▶ Near a solution, this bound controls the Hessian mismatch in the Newton error equation.

Local convergence of Newton's method

Theorem

Let f be twice continuously differentiable near x^* , with $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$. Suppose its Hessian is ρ -Lipschitz on a neighborhood of x^* . If x_0 is sufficiently close to x^* , Newton's iterates are well defined and satisfy

$$\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$$

for some constant $C > 0$ and every $k \geq 0$.

- ▶ This convergence is quadratic and therefore superlinear.
- ▶ Once inside the neighborhood, the error is squared at each step, so the number of correct digits roughly doubles.

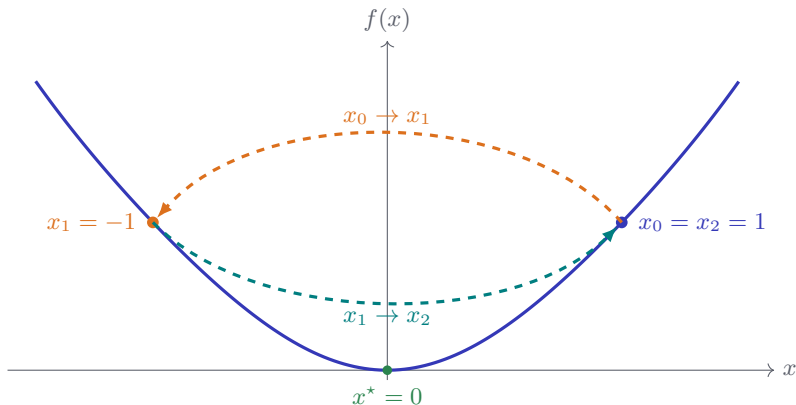
Note that:

- ▶ Each iteration requires Hessian information and a linear solve.
- ▶ The guarantee is local; the initial point x_0 has to be close enough to the optimal solution x^* .

Global behavior of Newton's method

Consider the following strongly convex objective:

$$f(x) = \frac{1}{2}x^2 + \frac{1 - \cos(\pi x)}{2\pi^2}.$$



- A Newton step can be too aggressive when away from the minimizer.

How far should we move along a search direction?

After choosing a search direction $p_k \in \mathbb{R}^d$, an iterative method updates by

$$x_{k+1} = x_k + \alpha_k p_k, \quad \alpha_k > 0.$$

Method	Usual choice	Limitation
Gradient descent	$\alpha_k = 1/L$	This guarantee requires global L -smoothness; L may be unknown, and $1/L$ may be overly conservative.
Newton's method	Full step $\alpha_k = 1$	The quadratic model is local, so a full step may overshoot far from a minimizer.

- Can we choose α_k adaptively using information available at the current iterate?

- ▶ After choosing a search direction p_k at the current iterate x_k , the step-size problem is one-dimensional.
- ▶ Consider the objective along the search line, for $\alpha \geq 0$:

$$\phi_k(\alpha) := f(x_k + \alpha p_k).$$

- ▶ How should we choose the step size α_k ?

Definition

Given the current iterate x_k and search direction p_k , choose a step size

$$\alpha_k^{\text{exact}} \in \arg \min_{\alpha \geq 0} f(x_k + \alpha p_k).$$

- ▶ Apply gradient descent with exact line search to $f(x) = \frac{1}{2}x^T Qx - b^T x$, where $Q \succ 0$. What is the update rule?

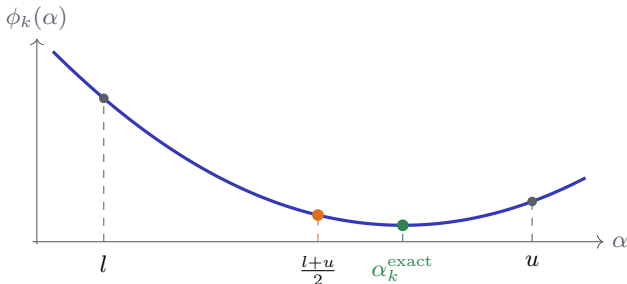
$$g_k := \nabla f(x_k) = Qx_k - b \neq 0, \quad \alpha_k^{\text{exact}} = \frac{g_k^T g_k}{g_k^T Q g_k}, \quad x_{k+1} = x_k - \alpha_k^{\text{exact}} g_k.$$

Exact line search

- ▶ For common objectives, there is no closed-form solution for exact line search.
- ▶ The exact line search requires solving the 1d problem

$$\min_{\alpha \geq 0} \phi_k(\alpha) := f(x_k + \alpha p_k).$$

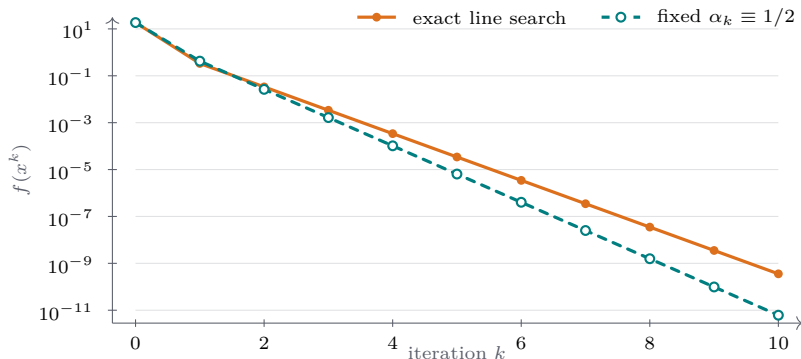
- ▶ If ϕ_k is convex, we can use a method such as bisection to find its minimizer.



- ▶ Solving the subproblem accurately may require many function evaluations.
- ▶ Exact line search is *one-step optimal*: it minimizes the next objective value along the current ray. It may be suboptimal in the long run.

Exact line search versus constant step size

$$f(x) = \frac{3}{4}x_1^2 + x_2^2 + 1 - \cos x_2, \quad x^0 = (3, \pi)^T, \quad p^k = -\nabla f(x^k).$$



- In this example, exact line search wins at the first step, but the constant step wins at later iterations.

Armijo condition – sufficient decrease

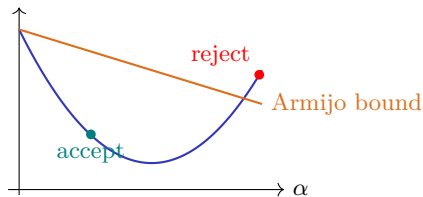
- ▶ Idea: Find the next iterate that lowers the objective value sufficiently.
- ▶ What do we mean by *sufficient*?
- ▶ Recall that p_k is a descent direction at x_k if $\nabla f(x_k)^T p_k < 0$.

Armijo condition

Fix $c_A \in (0, 1)$. A step size $\alpha > 0$ is acceptable if

$$f(x_k + \alpha p_k) \leq f(x_k) + c_A \alpha \nabla f(x_k)^T p_k.$$

- ▶ For a descent direction, the right-hand side is below $f(x_k)$.



Algorithm (Descent method with backtracking line search)

Require: $x_0 \in \mathbb{R}^d$, $\varepsilon \geq 0$, $\bar{\alpha} > 0$, and $c_A, \eta \in (0, 1)$

```
1: for  $k = 0, 1, 2, \dots$  do
2:   if  $\|\nabla f(x_k)\| \leq \varepsilon$  then return  $x_k$ 
3:   Choose  $p_k$  such that  $\nabla f(x_k)^T p_k < 0$ 
4:    $\alpha \leftarrow \bar{\alpha}$ 
5:   while  $f(x_k + \alpha p_k) > f(x_k) + c_A \alpha \nabla f(x_k)^T p_k$  do
6:      $\alpha \leftarrow \eta \alpha$ 
7:   end while
8:    $\alpha_k \leftarrow \alpha$ ,  $x_{k+1} \leftarrow x_k + \alpha_k p_k$ 
9: end for
```

Parameter	Role	Typical
ε	gradient stopping tolerance	problem-dependent
$\bar{\alpha}$	initial trial step	1
c_A	required decrease fraction	10^{-4}
η	shrink factor after rejection	1/2

- **Practical performance can vary substantially:** the line search parameters affect both accepted step sizes and function evaluations.

Will backtracking find an acceptable step?

- For $f(x) = \frac{1}{2}(x_1^2 + 4x_2^2)$ at $x = (2, 1)^T$, take $p = -\nabla f(x)$. Starting with $\bar{\alpha} = 1$, $\eta = 1/2$, and $c_A = 1/4$, which trial step is accepted first?

Here $f(x) = 4$, $p = (-2, -4)^T$, and $\nabla f(x)^T p = -20$.

α	$x + \alpha p$	$f(x + \alpha p)$	Armijo bound
1	$(0, -3)^T$	18	-1 reject
1/2	$(1, -1)^T$	5/2	3/2 reject
1/4	$(3/2, 0)^T$	9/8	11/4 accept

- The first accepted trial is $\alpha = 1/4$, giving $x_{\text{new}} = x + \alpha p = (3/2, 0)^T$.

Theorem

Let $\bar{\alpha} > 0$ and $c_A, \eta \in (0, 1)$. If f is continuously differentiable near x_k and p_k is a descent direction, then the backtracking loop terminates after finitely many reductions.

Theorem

Assume $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is bounded below and L -smooth for some $L > 0$, and every chosen direction satisfies, for fixed $\sigma, \beta > 0$,

$$\nabla f(x_k)^T p_k \leq -\sigma \|\nabla f(x_k)\|^2, \quad \|p_k\| \leq \beta \|\nabla f(x_k)\|.$$

Run the descent method with backtracking line search using $\varepsilon = 0$ and fixed $\bar{\alpha} > 0$ and $c_A, \eta \in (0, 1)$. Set $f^* := \inf_x f(x)$ and $\underline{\alpha} := \min\{\bar{\alpha}, 2\eta(1 - c_A)\sigma/(L\beta^2)\} > 0$. For every $k \geq 1$,

$$\min_{0 \leq j < k} \|\nabla f(x_j)\|^2 \leq \frac{f(x_0) - f^*}{c_A \sigma \underline{\alpha} k}.$$

Moreover, $\|\nabla f(x_k)\| \rightarrow 0$.

- The constants σ and β are uniform in k ; strict descent without these quantitative safeguards is insufficient.

Corollary

Assume $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is twice continuously differentiable, μ -strongly convex, and L -smooth. Run Newton's method with backtracking line search from any x_0 , using $\varepsilon = 0$, the direction

$$p_k = -\nabla^2 f(x_k)^{-1} \nabla f(x_k)$$

and fixed $\bar{\alpha} > 0$ and $c_A, \eta \in (0, 1)$. Then $x_k \rightarrow x^*$, where x^* is the unique minimizer of f .

- ▶ The Hessian bounds give $\sigma = 1/L$ and $\beta = 1/\mu$ in the theorem, so $\|\nabla f(x_k)\| \rightarrow 0$. Strong convexity then gives $\|x_k - x^*\| \leq \|\nabla f(x_k)\| / \mu$.
- ▶ If $\bar{\alpha} = 1$, $c_A < 1/2$, and the Hessian is Lipschitz near x^* , full steps are eventually accepted and convergence becomes quadratic.

Summary: gradient descent

- ▶ Run gradient descent with $\varepsilon = 0$ and $\alpha_k = 1/L$, where f is L -smooth for some $L > 0$.

Additional assumption	Guarantee for every $k \geq 1$
-----------------------	--------------------------------

f is bounded below, with $f^* := \inf_x f(x)$	$\min_{0 \leq j < k} \ \nabla f(x_j)\ ^2 \leq \frac{2L(f(x_0) - f^*)}{k}$
--	---

f is convex and x^* is a minimizer	$f(x_k) - f(x^*) \leq \frac{L \ x_0 - x^*\ ^2}{2k}$
---	---

f is μ -strongly convex	$f(x_k) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^k (f(x_0) - f(x^*))$
-------------------------------	---

- ▶ More geometric structure strengthens the guarantee: best-iterate stationarity, then sublinear and linear objective convergence.

Summary: Newton's method and line search

- ▶ **One-step solution for quadratics.** For $f(x) = \frac{1}{2}x^T Qx - b^T x$ with $Q \succ 0$, full-step Newton reaches $x^* = Q^{-1}b$ from any starting point in one iteration.
- ▶ **Local quadratic convergence.** If $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$, and the Hessian is Lipschitz near x^* , every sufficiently close start satisfies $\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$.
- ▶ **Finite backtracking.** If f is continuously differentiable near x_k and p_k is a descent direction, Armijo backtracking accepts a step after finitely many reductions.
- ▶ **Global stationarity.** For an L -smooth, bounded-below objective, the uniform direction safeguards ensure that descent with backtracking and $\varepsilon = 0$ satisfies $\|\nabla f(x_k)\| \rightarrow 0$.
- ▶ **Globalized Newton.** If $f \in C^2$ is μ -strongly convex and L -smooth, Newton with backtracking converges from every x_0 to the unique minimizer. If also $\bar{\alpha} = 1$, $c_A < 1/2$, and the Hessian is locally Lipschitz, full steps are eventually accepted and convergence becomes quadratic.